

17-803 Empirical Methods

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Interviewing

Tuesday, February 3, 2026

Outline for Today

- ▶ Methodological details of interviews:
 - ▶ Setting up
 - ▶ Conducting
- ▶ In-class activity to design a protocol

Credits

▶ Graphics:

- ▶ Dave DiCello photography (cover)

▶ Content:

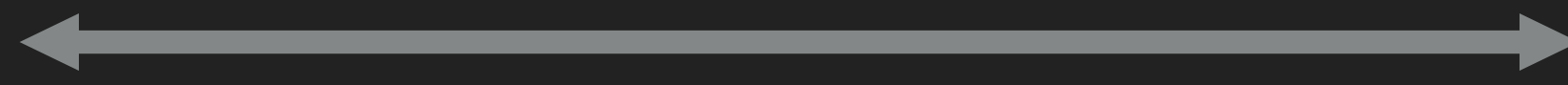
- ▶ King, N. (2004). **Using interviews in qualitative research**. In C. Cassell & G. Symon (Eds.), Essential Guide to Qualitative Methods in Organizational Research (pp. 11-22). London: Sage.
- ▶ Seidman, I. (2012). **Interviewing as qualitative research: A guide for researchers in education and the social sciences**: Teachers college press. (Ch 4, Ch 6).
- ▶ Goodman, P. (2005). **Building Effective Interviewing Skills**. CMU Tepper 70-311
- ▶ Weiss, R. S. (1995). **Learning from strangers: The art and method of qualitative interview studies**. Simon and Schuster. (Ch 4).
- ▶ Chris Bird's (2016) interview guide. Published in Perspectives on Data Science for Software Engineering <http://www.cabird.com/wp/bird2016interviews/>

Interviews are (the most) common method of data gathering in qualitative research.

A Variety of Forms of Qualitative Research Interviews

Questions

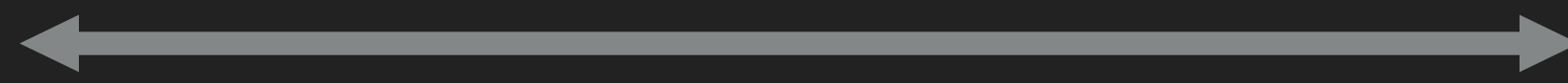
Specific



General

Order

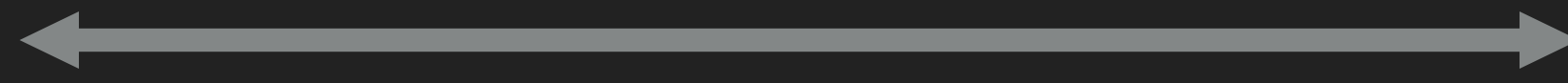
Predetermined / Fixed



Evolving / Flexible

Responses

Precise



Open ended

You Will Probably Encounter / Do Semi-Structured Interviews



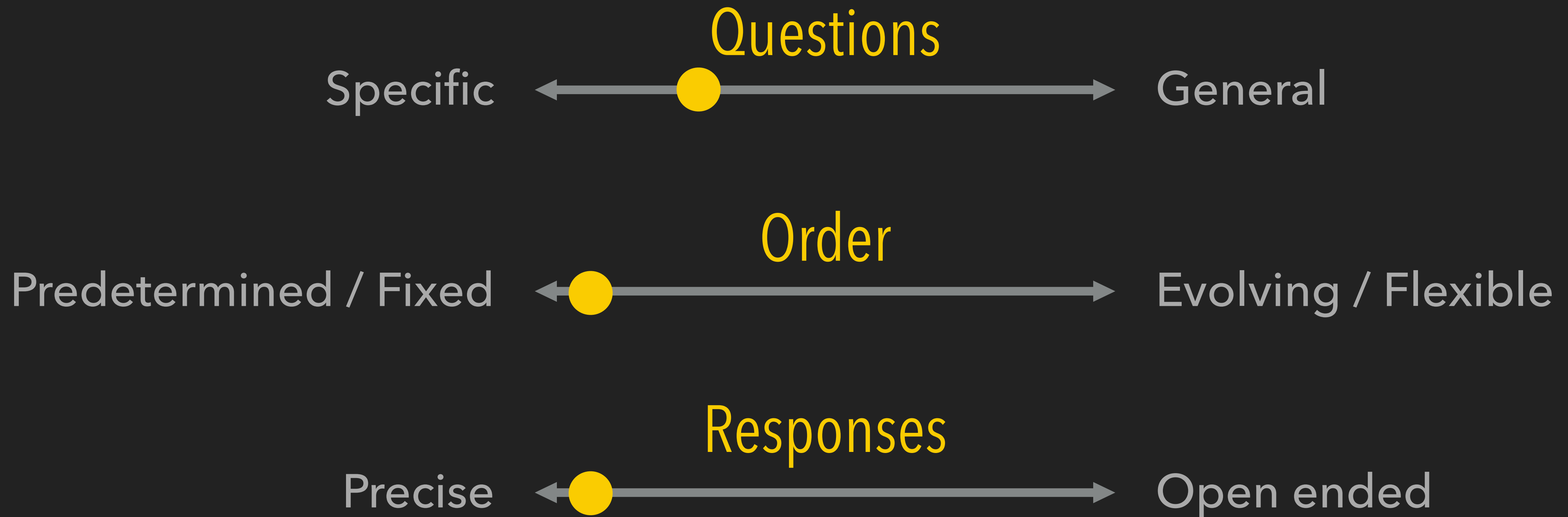
When To Consider Doing Interviews

- ▶ Wonderful tools for exploratory investigation.
- ▶ Often can drive the formation of theories and hypotheses.
 - ▶ common in mixed-methods designs
- ▶ Great way to validate data!

Chris Bird @Microsoft Research

- ▶ “I was investigating code review latency at Microsoft when I found something odd in the data for a team in Bing: Many of their code reviews were signed off in just minutes (sometimes under a minute) after the code review was created.
- ▶ “I meticulously looked at the data collection code, I conducted a number of statistical tests on the data based on guesses that I had. Nothing.
- ▶ “As a last resort, I contacted one of the developers on the team and scheduled an interview. She explained that the reason for the lightning fast reviews was that they often conduct code reviews in person with two or three reviewers huddled around the developer's screen as they explain the change. Once the reviewers were happy, the author would create the request in the review system and the reviewers would immediately sign off on the review.
- ▶ “In just a few minutes, she had answered a question that I hadn't been able to answer after hours of testing hypotheses on data. You can learn things in an interview that you would never have thought of yourself.”

Aside: You Can Use Interviews in Quantitative Research



Aside: You Can Use Interviews in Quantitative Research

- ▶ **Quant:** Interviewee as research *subject*
- ▶ **Qual:** Interviewee as research *participant*
 - ▶ actively shaping the course of the interview rather than passively responding to the interviewer's pre-set questions

Main Goals and Characteristics of Qualitative Interviews

▶ Goal:

- ▶ see the research topic from the **perspective of the interviewee**;
- ▶ understand **how and why** they come to have this particular perspective.

▶ Characteristics:

- ▶ **low degree of structure** imposed by the interviewer;
- ▶ preponderance of **open questions**;
- ▶ **focus on specific situations and action sequences** in the world of the interviewee rather than abstractions and general opinions.

General Pros

- ▶ Interviews allow **rich engagement and follow up** questions.
- ▶ Collect historical **data that is not recorded anywhere**.
- ▶ Elicit opinions and impressions in **richer detail** than people would provide through written communication.
- ▶ Information from interviews **can be triangulated** with other data sources.
- ▶ Interviews **can be used to clarify things** that have already happened (especially following an observation).

General Cons

- ▶ Usually small **sample size**.
- ▶ The **time** required for each individual interview.
- ▶ The challenge of **finding appropriate interviewees and scheduling** a time that works for all parties.
- ▶ Potential **bias introduced by the interviewer** during the interview (word choice, tone of voice, body language can all affect responses).
- ▶ The **time** required for transcription and subsequent analysis.

Steps

1. Defining the research question

- ▶ Typical focus is on how participants describe and make sense of particular elements of their lives.
- ▶ Goal is *not* to quantify individual experience.
- ▶ Avoid reflecting your own presuppositions or biases.

2. Creating the interview guide (protocol)

3. Recruiting participants

4. Carrying out the interviews

5. Analyzing the data

~~Step 1: Defining the research question~~
(We talked about this before)

~~Step 5: Analyzing the data~~
(Next time)

Step 2: Creating the interview guide

Interview Guide

- ▶ **Not formal schedule** of questions to be asked word-for-word in a set order
- ▶ Instead, list:
 - ▶ **topics** the interviewer should attempt to cover
 - ▶ **probes** which may be used to follow-up responses and elicit greater detail from participants
- ▶ **Guide can evolve** after each interview:
 - ▶ adding probes / topics that emerged spontaneously in interviews
 - ▶ dropping or re-formulating those which are incomprehensible or consistently fail to elicit relevant responses

General Topics To Be Covered: From Open to More Specific, Then Back To Open

- ▶ Introduction
- ▶ Background / History
- ▶ Opening questions
 - ▶ e.g., satisfactions, frustrations
- ▶ Follow-up questions
 - ▶ More detail
 - ▶ Specific areas to elaborate on
- ▶ Other important points
 - ▶ "Anything to add?"
 - ▶ "Anything we missed?"
- ▶ Wrap-up
 - ▶ Express appreciation



Typical Practice

- ▶ Include many information-seeking questions
 - ▶ followed up with probes to explore the interviewees' views and experiences in more depth
- ▶ Prefer questions that focus on concrete examples, rather than abstracted generalities
- ▶ Incorporates fully formed questions as well as just topic headings
 - ▶ The latter encourage the interviewer to be responsive to the interviewee and avoid presuppositions

Step 3: Recruiting participants

Considerations

- ▶ Amount of time and resource available
- ▶ Diversity of expected views
 - ▶ Representativeness of the sample
 - ▶ Depends on the design for which the interviews are being used

Purposeful Sampling Techniques

- ▶ Typical case
- ▶ Extreme or deviant case
- ▶ Critical case
- ▶ Sensitive case
- ▶ Convenience
- ▶ **Maximum variation**
 - ▶ Preferred, should allow the widest possibility for readers to connect to what they are reading

How Many Participants Are Enough?

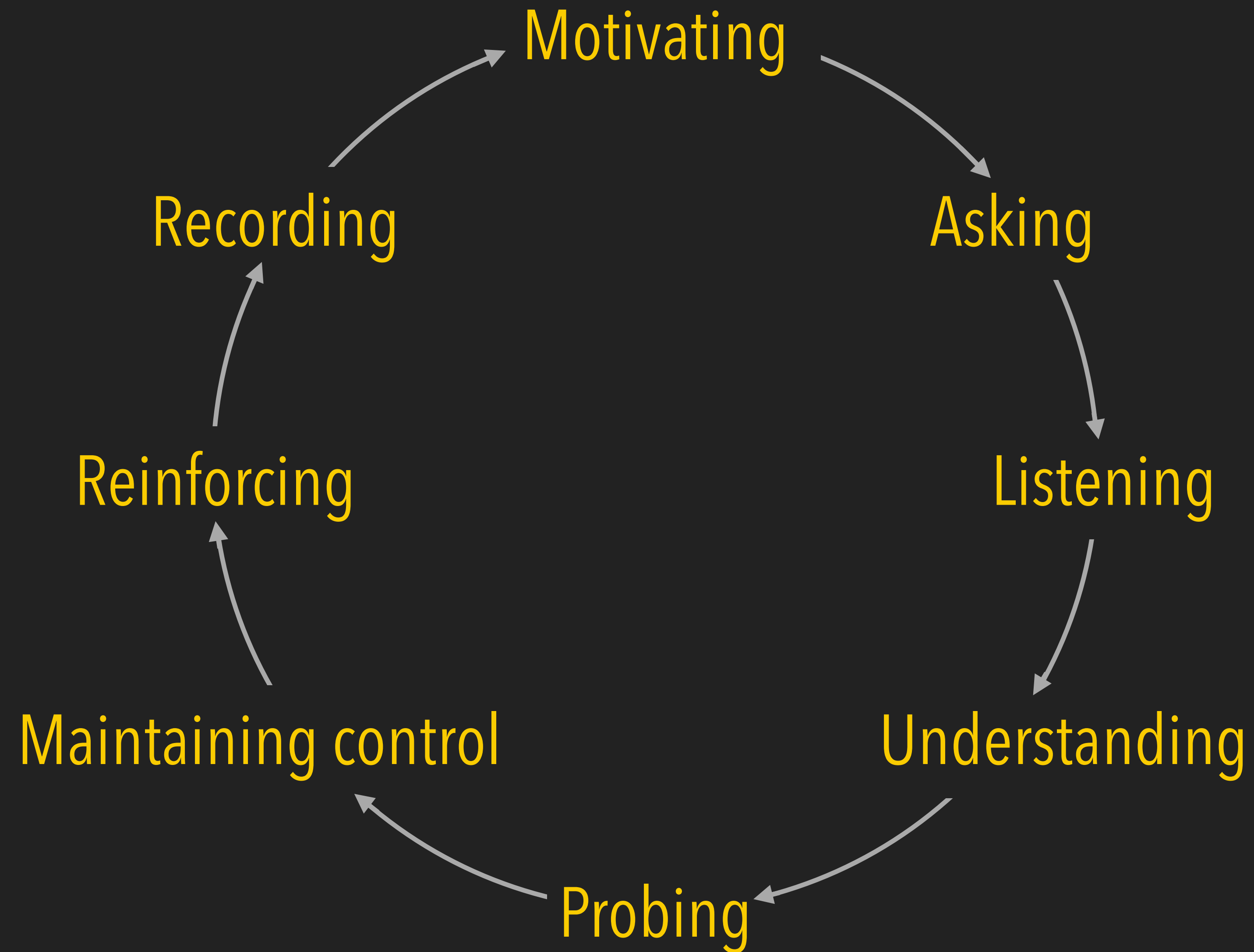
- ▶ Sufficiency
 - ▶ reflect the range of participants and sites that make up the population
- ▶ Saturation of information
 - ▶ not learning anything new
- ▶ Remember, you need to satisfy reviewers as well as yourself!

Considerations (2)

- ▶ **Access** through a gatekeeper
 - ▶ Formal or informal
 - ▶ Access through peers preferable to “above” / “below”
- ▶ **Avoid interviewing “easy access” subjects**
 - ▶ Need to have enough distance from interviewees so that they take nothing for granted
- ▶ **Confidentiality**

Step 4: Conducting the interviews

Interview Process



Interview Process



Motivating the Interviewee To Participate

- ▶ The purpose of the interview should be **related to the respondent's own goals and values**.
- ▶ Explain **what will be expected** of the interviewee in the course of the interview (e.g., the length of the interview, degree of expertise required).
- ▶ Specify the way in which information is used.

Motivating the Interviewee To Produce

- ▶ The interviewer must create and maintain an atmosphere in which the respondent feels **fully understood** and **safe to communicate fully** without fear of being judged or criticized.
- ▶ Provide periodic **reinforcement** on the process of the interview.

Motivating — Have Answers Ready for:

- ▶ Why are you doing this study?
- ▶ What do you get out of it?
- ▶ What does university X / company Y get out of this?
- ▶ What will I get out of this?
- ▶ Who is paying for this?
- ▶ Will I receive the data?
- ▶ How do I know it will be confidential?
- ▶ How long will this take?
- ▶ Are you doing other parts of the company?
- ▶ Why did you select my group / me?

Interview Process



Asking Questions

- ▶ Avoid asking **multiple questions** at once:
 - ▶ 'Why did you join this open-source project, and do you think it has brought benefits to your programming experience?'
- ▶ Avoid **leading questions**:
 - ▶ 'So you felt that using NL2Code improved your productivity?'
 - ▶ Instead: 'What, if any, impact did NL2Code have on your productivity?'
 - ▶ 'Your parents pushed you to study, didn't they?'
 - ▶ 'How satisfied were you with NL2Code?'

Asking Questions

- ▶ Avoid assuming that the answer to a question is so obvious that it need not be asked:
 - ▶ 'How concerned are you about your privacy online?'
- ▶ Avoid imposing your perception:
 - ▶ 'So what you're really saying is ...'

Asking Questions

- ▶ Avoid ending the interview on a difficult, threatening or painful topic.
 - ▶ Instead, finish by giving the interviewee the opportunity to make any comments about the subject at hand which have not been covered in the rest of the interview:
 - ▶ 'What else, if anything, should I have asked?'

Asking Questions

- ▶ Do ask questions in a simple, direct, clear manner.
- ▶ Do be flexible:
 - ▶ Topic order may change during interview
- ▶ Do open with a question which can be answered easily and without potential embarrassment or distress
 - ▶ E.g., requests for factual or descriptive information

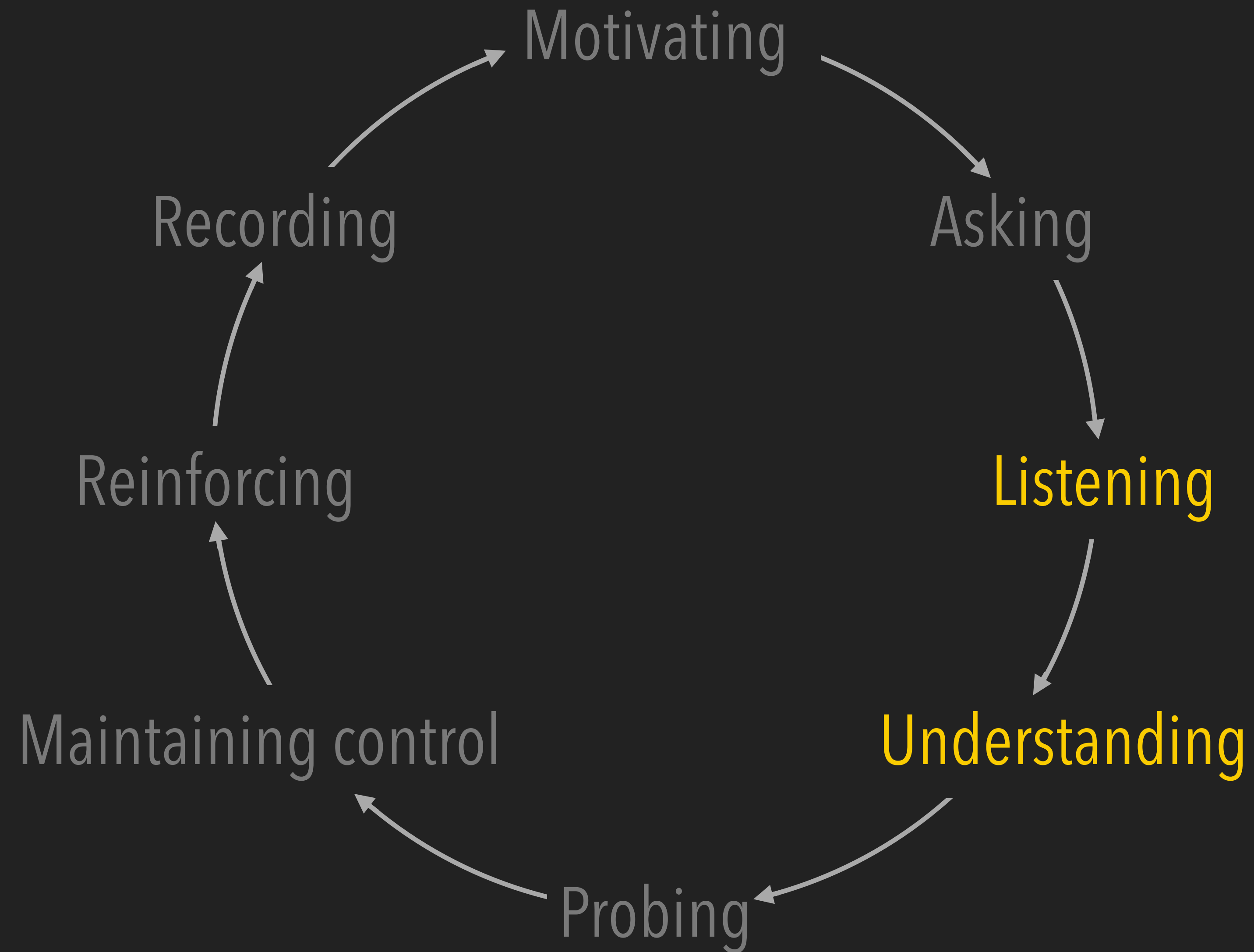
Asking Questions

- ▶ Do ask open ended questions
 - ▶ “Grand tour” questions:
 - ▶ ‘Take me through a day in your work life.’
 - ▶ ‘Reconstruct your day for me from the time you wake up to the time you go to bed.’
 - ▶ Subjective experience questions:
 - ▶ ‘What was attending this class like for you?’

Asking Questions

- ▶ Do follow up on what the participant says.
 - ▶ Clarifications, details, stories
 - ▶ Trust your instincts, explore emerging directions
- ▶ Do ask participants to reconstruct, not remember, their experience:
 - ▶ 'What happened?' (reconstruction) vs
 - ▶ 'Do you remember what happened?' (memory)
- ▶ Do ask for concrete details.

Interview Process



Listening Means

- ▶ Attending to verbal and non-verbal cues
- ▶ Attending to the total message – facts and feelings
- ▶ Being an active listener
- ▶ Testing listening with feedback

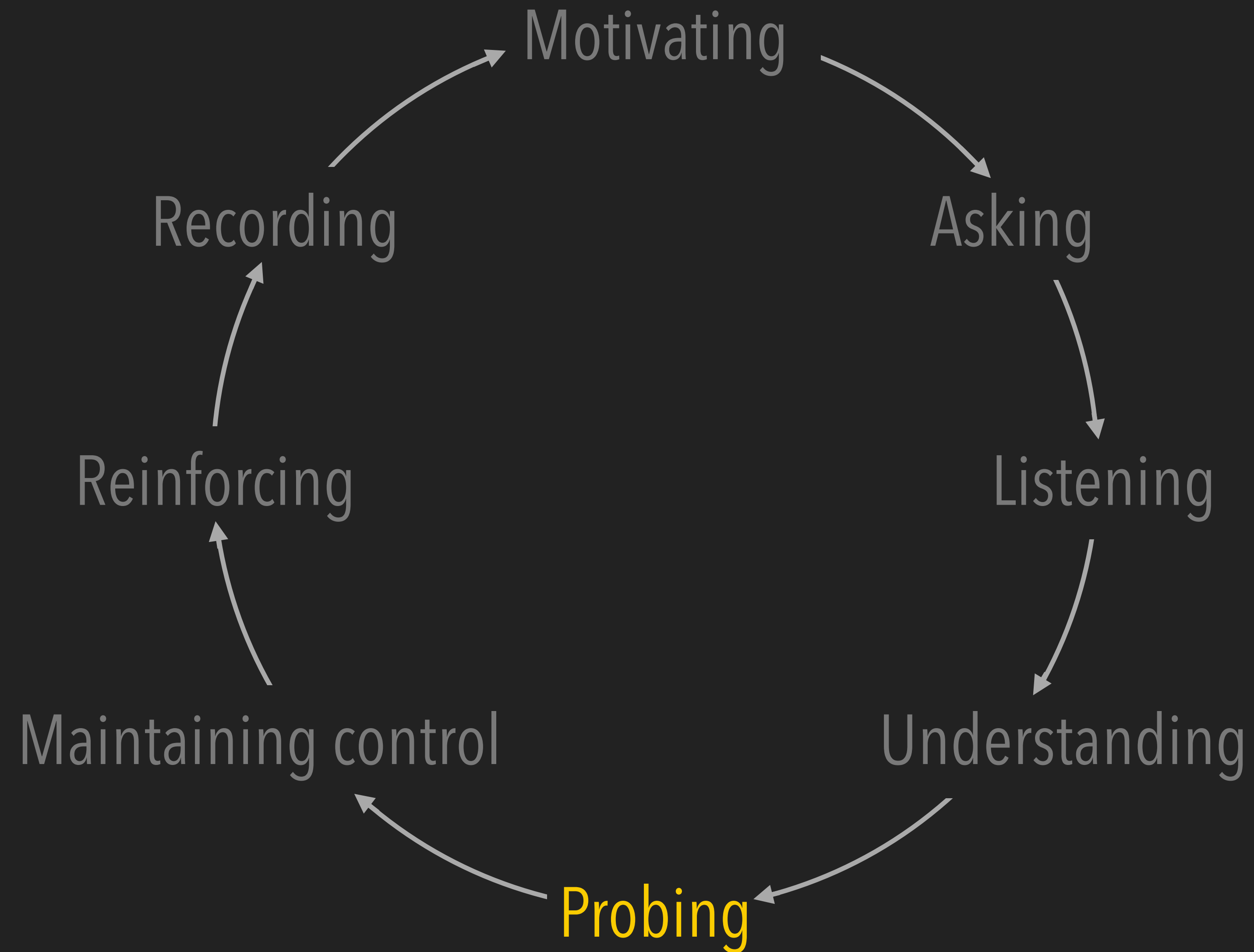
Understanding Means

- ▶ Put yourself in the interviewee's frame of reference
- ▶ Be non-evaluative
- ▶ Don't prematurely analyze or draw conclusions

Listen More, Talk Less!

- ▶ Three levels:
 - ▶ The **substance**, must internalize what participants say
 - ▶ “**Inner voice**” vs “**outer voice**” (e.g., ‘challenge’, ‘adventure’, ‘fascinate’)
 - ▶ The **process** (assess progress, stay alert for cues about how to move the interview forward)
- ▶ Control mechanism:
 - ▶ listen to yourself in recordings / check length of your paragraphs in transcripts

Interview Process



Directive Probes

- ▶ **Open-ended probe:**

- ▶ You: "What were the major responsibilities of your most recent job?"

- ▶ **Specific probe:**

- ▶ Them: "I've always had the ability to learn a new programming language quickly."
 - ▶ You: "What specific steps do you take to learn a new language?"
- ▶ You: "How would you rate your contributions to this open source project?"
 - ▶ Them: "I think I'm a major contributor"
 - ▶ You: "I'm glad to hear that. What contributions in particular made you feel that way?"

Directive Probes

- ▶ **Bi-polar probe:**

- ▶ You: "Most people I talked to can identify aspects of working remotely they really like and other aspects of working remotely that they dislike. Would you tell me about those aspects that you really like?"
 - ▶ then repeat question using dislike

- ▶ **Elaboration probe:**

- ▶ You: "Is there anything else?"
- ▶ You: "What else can you think about?"
- ▶ You: "Any other thoughts?"

Non-Directive Probes

- ▶ Reflecting feelings:

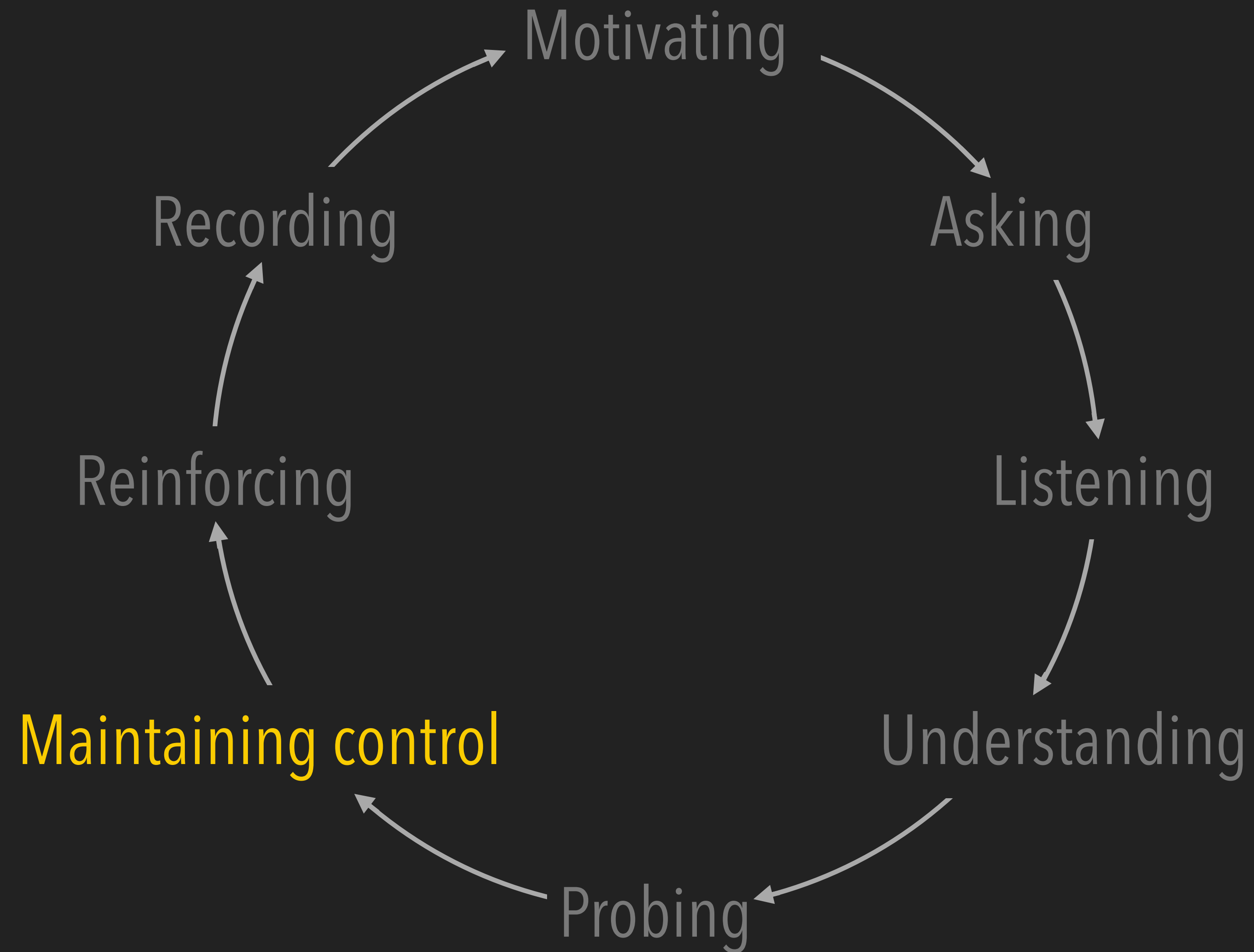
- ▶ Them: "...I've been here for 15 years and I don't feel I have been treated fairly."
- ▶ You: "You feel you haven't been treated fairly?"

- ▶ Indirect follow-ups:

- ▶ You: "...tell me more"
- ▶ You: "I would like to hear more about that point, could you elaborate a bit?"

- ▶ Pause

Interview Process



Managing the Process

- ▶ Recall – active listening
 - ▶ assess progress, stay alert for cues about how to move the interview forward
- ▶ Follow up, but don't interrupt
 - ▶ Instead, take notes and return:
 - ▶ "A while back you talked about X. Would you talk more about that X?"

Dealing With Difficult Interviewees

- ▶ **Uncommunicative interviewee** – symptom: monosyllabic answers
 - ▶ Helps to be clear about required time and anonymity
 - ▶ Phrase questions as open as possible
 - ▶ Use silence
- ▶ **Over-communicative interviewee** – symptom: repeatedly straying far from your questions without adding anything of significant interest
 - ▶ Politely interrupt at a natural pause; refer back to an earlier point:
 - ▶ “That’s very interesting. Could we go back to what you were saying earlier about X, I’d like you to tell me more about that”

Dealing With Difficult Interviewees

▶ High-status interviewees:

- ▶ Don't appear over-familiar
- ▶ Nor overly nervous or submissive
- ▶ Instead, be respectful but confident of the worth of what you are doing and of your own expertise.

▶ Emotionally charged topics:

- ▶ Give people the time they require to answer your questions
- ▶ Avoid non-verbal cues that might be taken as indicating impatience
- ▶ Skip question, return later if possible

Interview Process



Feedback / Reinforcement

- ▶ Interviewees need to know how they are doing in producing information.
- ▶ Reinforce early and often (i.e. throughout the interview).
- ▶ Provide positive feedback without reinforcing specific content:
 - ▶ "We are making good progress."
 - ▶ "I appreciate your willingness to help us in this project."
 - ▶ "Your comments are very helpful to us."

Dealing With “Contradictions”

- ▶ What you consider a contradiction may not be for the respondent
- ▶ Note any perceived contradictions with a quick note
- ▶ Postpone probing contradictions
- ▶ Wait for some natural break
- ▶ Feed back the ideas that seem to be conflicting in a neutral way:
 - ▶ “You mentioned X. You also mentioned Y. Help me understand the relationship between these two points.”

Interview Process



Recordings and Transcripts

- ▶ Record the interview, capture everything verbatim.
- ▶ But still take copious notes during the interview.
 - ▶ Much easier to refer to notes than find a particular place in a recording.
- ▶ Whatever can go wrong, will go wrong!

Activity: Workshop an Interview Guide (1/2)

- ▶ Goal: Translate the knowledge gaps identified in your literature review into an interview study designed to uncover how researchers are actually integrating GenAI and agentic tools into their workflows.
- ▶ Look at the open research question you wrote for your homework. As a group, select one (or synthesize a new one) that is best suited for an interview-based approach.
- ▶ Decide on your sampling frame. Who are you talking to?
 - ▶ Are you looking for extreme users (early adopters in SCS) vs. skeptics?
 - ▶ Are you interviewing PIs (who set policy) or PhDs (who do the “hands-on” prompting)?
 - ▶ Define your inclusion/exclusion criteria.

Activity: Workshop an Interview Guide (2/2)

- ▶ Draft a semi-structured interview guide intended for a 20-30 minute session. Include:
 - ▶ A neutral opening that encourages honesty (avoiding social desirability bias where they might hide AI use).
 - ▶ Questions specifically targeting the use of agentic tools (tools that act on their own) vs. simple chatbots.
 - ▶ Questions designed to see if GenAI is changing their actual research questions, not just their speed.
- ▶ Think about how you move beyond the informant saying "I trust it" to understanding the mechanisms of that trust.

Sample Interview Protocol: Agentic GenAI in the Research Workflow

- ▶ To start, think of a specific research project you are working on right now. In the last week, which GenAI tools—if any—have been open in your browser or running in your terminal?
 - ▶ Are these tools part of your formal 'official' workflow, or are they more like an 'under-the-desk' assistant?
- ▶ We often talk about LLMs as chatbots, but I'm interested in agents—tools that you set a goal for and let run (e.g., an automated literature mapper or a code-generating loop).
- ▶ Can you describe a task where you've felt comfortable 'handing over the keys' to an AI tool to let it work autonomously for a period of time?
 - ▶ What is a research task you would never delegate to an agent right now? Why is that the boundary?"
- ▶ Recall a specific instance where an AI tool provided an output—a summary, a piece of code, or a data visualization—that looked correct but felt 'off.' Walk me through how you verified it.
 - ▶ At what point did you stop checking? What was the 'signal' that told you it was now safe to use in your paper or analysis?
- ▶ Do you feel the standard for 'verifying' AI output in your specific lab/department is higher or lower than what you see in the general literature?
- ▶ Some argue GenAI just makes us faster. Others say it changes what we ask. Since you've started using these tools, have you pursued a research lead or a 'what if' question that you would have previously ignored because it was too labor-intensive?
- ▶ If you had to mentor a first-year PhD student starting today, what is the one 'GenAI habit' you would tell them to avoid to protect the quality of their science?